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Python Interview Questions

w i t h a n s w e r s



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Basic Level



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1. What is the Difference Between a Shallow Copy and Deep Copy?

Answer:

Deepcopy creates a different object and populates it with the child objects of the original object. Therefore, changes in the original object are not reflected in the copy.

`copy.deepcopy()` creates a Deep Copy.

Shallow copy creates a different object and populates it with the references of the child objects within the original object. Therefore, changes in the original object are reflected in the copy.

`copy.copy` creates a Shallow Copy.

2. How Is Multithreading Achieved in Python?

Answer:

Multithreading usually implies that multiple threads are executed concurrently. The Python Global Interpreter Lock doesn't allow more than one thread to hold the Python interpreter at that particular point of time. So multithreading in python is achieved through context switching. It is quite different from multiprocessing which actually opens up multiple processes across multiple threads.

3. Discuss Django Architecture.

Answer:

Django is a web service used to build your web pages. Its architecture is as shown:

- Template: the front end of the web page
- Model: the back end where the data is stored
- View: It interacts with the model and template and maps it to the URL
- Django: serves the page to the user



4. What Advantage Does the Numpy Array Have over a Nested List?

Answer:

Numpy is written in C so that all its complexities are backed into a simple to use a module. Lists, on the other hand, are dynamically typed. Therefore, Python must check the data type of each element every time it uses it. This makes Numpy arrays much faster than lists.

Numpy has a lot of additional functionality that list doesn't offer; for instance, a lot of things can be automated in Numpy.

5. What are Pickling and Unpickling?

Answer:

Pickling	Unpickling
Converting a Python object hierarchy to a byte stream is called pickling. Pickling is also referred to as serialization.	Converting a byte stream to a Python object hierarchy is called unpickling. Unpickling is also referred to as deserialization.

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If you just created a neural network model, you can save that model to your hard drive, pickle it, and then unpickle to bring it back into another software program or to use it at a later time.

6. How is Memory managed in Python?

Answer:

Python has a private heap space that stores all the objects. The Python memory manager regulates various aspects of this heap, such as sharing, caching, segmentation, and allocation. The user has no control over the heap; only the Python interpreter has access.

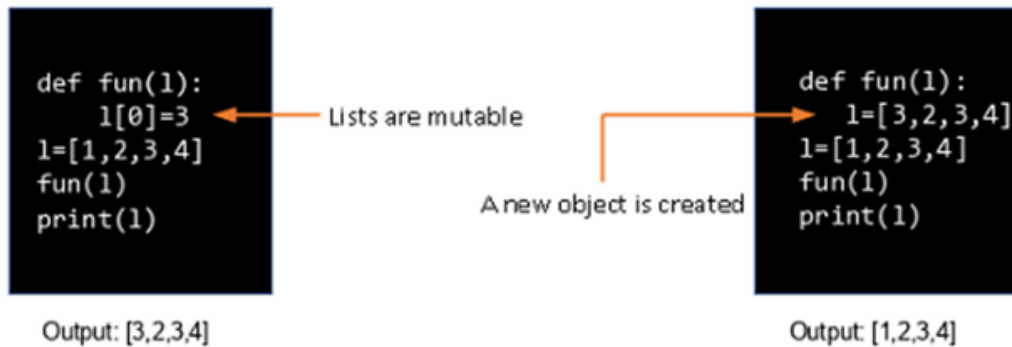
7. Are Arguments in Python Passed by Value or by Reference?

Answer:

Arguments are passed in python by a reference. This means that any changes made within a function are reflected in the original object. Consider two sets of code shown below:

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In the first example, we only assigned a value to one element of 'l', so the output is [3, 2, 3, 4].

In the second example, we have created a whole new object for 'l'. But, the values [3, 2, 3, 4] doesn't show up in the output as it is outside the definition of the function.

8. How Would You Generate Random Numbers in Python?

Answer:

To generate random numbers in Python, you must first import the random module.

The random() function generates a random float value between 0 & 1.

> random.random()

The randrange() function generates a random number within a given range.

Syntax: randrange(beginning, end, step)

Example - > random.randrange(1,10,2)

9. What Does the // Operator Do?

Answer:

In Python, the / operator performs division and returns the quotient in the float.

For example: 5 / 2 returns 2.5

The // operator, on the other hand, returns the quotient in integer.

For example: 5 // 2 returns 2

10. What Does the 'is' Operator Do?



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Answer:

The 'is' operator compares the id of the two objects.

```
list1=[1,2,3]
```

```
list2=[1,2,3]
```

```
list3=list1
```

```
list1 == list2    True
```

```
list1 is list2    False
```

```
list1 is list3    True
```

11. What Is the Purpose of the Pass Statement?

Answer:

The pass statement is used when there's a syntactic but not an operational requirement.

12. How Will You Check If All the Characters in a String are Alphanumeric?

Answer:

Python has an inbuilt method `isalnum()` which returns true if all characters in the string are alphanumeric.

Example -

```
>> "abcd123".isalnum()
```

```
Output: True
```

```
>> "abcd@123#".isalnum()
```

```
Output: False
```

Another way is to use regex as shown.

```
>> import re
```

```
>> bool(re.match('[A-Za-z0-9]+$', 'abcd123'))
```

```
Output: True
```

```
>> bool(re.match('[A-Za-z0-9]+$', 'abcd@123'))
```

```
Output: False
```

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13. How Will You Merge Elements in a Sequence?

Answer:

There are three types of sequences in Python:

- Lists
- Tuples
- Strings

Example of Lists -

```
>>l1=[1,2,3]
>>l2=[4,5,6]
>>l1+l2
```

Output: [1,2,3,4,5,6]

Example of Tuples -

```
>>t1=(1,2,3)
>>t2=(4,5,6)
>>t1+t2
```

Output: (1,2,3,4,5,6)

Example of String -

```
>>s1="Talent"
>>s2="Battle"
>>s1+s2
```

Output: 'TalentBattle'

13. How Will You Merge Elements in a Sequence?

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Example of Lists -

```
>>l1=[1,2,3]
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Output: [1,2,3,4,5,6]

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```

Output: (1,2,3,4,5,6)

Example of String -

```
>>s1="Talent"
>>s2="Battle"
>>s1+s2
```

Output: 'TalentBattle'

14. How Would You Remove All Leading Whitespace in a String?

Answer:

Python provides the inbuilt function `lstrip()` to remove all leading spaces from a string.

```
>> " Python".lstrip
```

Output: Python

15. How Would You Replace All Occurrences of a Substring with a New String?

Answer:

The `replace()` function can be used with strings for replacing a substring with a given string. Syntax:

```
str.replace(old, new, count)
```

`replace()` returns a new string without modifying the original string.

Example -

```
>> "Hey John. How are you, John?".replace("john", "John", 1)
```

Output: "Hey John. How are you, John?"

16. What Is the Difference Between `Del` and `Remove()` on Lists?

Answer:

The `replace()` function can be used with strings for replacing a substring with a given string. Syntax:

```
str.replace(old, new, count)
```

`replace()` returns a new string without modifying the original string.

Example -

```
>> "Hey John. How are you, John?".replace("john", "John", 1)
```

Output: "Hey John. How are you, John?"

del	remove()
• del removes all elements of a list within a given range • Syntax: <code>del list[start:end]</code>	• remove() removes the first occurrence of a particular character • Syntax: <code>list.remove(element)</code>



Here is an example to understand the two statements -

```
>>lis=['a', 'b', 'c', 'd']
```

```
>>del lis[1:3]
```

```
>>lis
```

```
Output: ["a","d"]
```

```
>>lis=['a', 'b', 'b', 'd']
```

```
>>lis.remove('b')
```

```
>>lis
```

```
Output: ['a', 'b', 'd']
```

Note that in the range 1:3, the elements are counted up to 2 and not 3.

17. How Do You Display the Contents of a Text File in Reverse Order?

Answer:

You can display the contents of a text file in reverse order using the following steps:

- Open the file using the open() function
- Store the contents of the file into a list
- Reverse the contents of the list
- Run a for loop to iterate through the list

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18. Differentiate Between `append()` and `extend()`.

Answer:

<code>append()</code>	<code>extend()</code>
• <code>append()</code> adds an element to the end of the list • Example - <pre>>>lst=[1,2,3] >>lst.append(4) >>lst Output:[1,2,3,4]</pre>	• <code>extend()</code> adds elements from an iterable to the end of the list • Example - <pre>>>lst=[1,2,3] >>lst.extend([4,5,6]) >>lst Output:[1,2,3,4,5,6]</pre>

19. What Is the Output of the below Code? Justify Your

Answer:

```
>>def addToList(val, list=[]):
>> list.append(val)
>> return list
>>list1 = addToList(1)
>>list2 = addToList(123,[])
>>list3 = addToList('a')
>>print ("list1 = %s" % list1)
>>print ("list2 = %s" % list2)
>>print ("list3 = %s" % list3)
```

Output:

```
list1 = [1,'a']
list2 = [123]
list3 = [1,'a']
```

Note that list1 and list3 are equal. When we passed the information to the addToList, we did it without a second value. If we don't have an empty list as the second value, it will start off with an empty list, which we then append. For list2, we appended the value to an empty list, so its value becomes [123]. For list3, we're adding 'a' to the list. Because we didn't designate the list, it is a shared value. It means the list doesn't reset and we get its value as [1, 'a']. Remember that a default list is created only once during the function and not during its call number.

20. What Is the Difference Between a List and a Tuple?

Answer:

Lists are mutable while tuples are immutable.

Example:

List

```
>>lst = [1,2,3]
```

```
>>lst[2] = 4
```

```
>>lst
```

Output:[1,2,4]

Tuple

```
>>tpl = (1,2,3)
```

```
>>tpl[2] = 4
```

```
>>tpl
```

Output:TypeError: 'tuple' the object does not support item assignment

There is an error because you can't change the tuple 1 2 3 into 1 2 4. You have to completely reassign tuple to a new value.

21. What Is Docstring in Python?

Answer:

Docstrings are used in providing documentation to various Python modules, classes, functions, and methods.

Example -

```
def add(a,b):
```

```
    """This function adds two numbers."""
```

```
    sum=a+b
```

```
    return sum
```

```
sum=add(10,20)
```

```
print("Accessing docstring method 1:",add.__doc__)
```

```
print("Accessing docstring method 2:",end="")
```

```
help(add)
```



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Output -

Accessing docstring method 1: This function adds two numbers.

Accessing docstring method 2: Help on function add-in module __main__:

add(a, b)

This function adds two numbers.

22. How Do You Use Print() Without the Newline?

Answer:

The solution to this depends on the Python version you are using.

Python v2

```
>>print("Hi. "),
```

```
>>print("How are you?")
```

Output: Hi. How are you?

Python v3

```
>>print("Hi",end=" ")
```

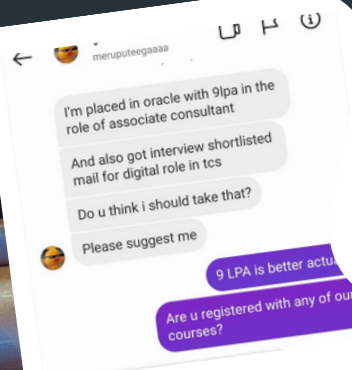
```
>>print("How are you?")
```

Output: Hi. How are you?

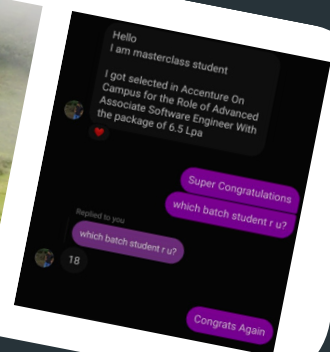
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25. Write a Function Prototype That Takes a Variable Number of Arguments

Answer:

The function prototype is as follows:

```
def function_name(*list)
>>def fun(*var):
>> for i in var:
print(i)
>>fun(1)
>>fun(1,25,6)
```

In the above code, * indicates that there are multiple arguments of a variable.

26. What Are *args and *kwargs?

Answer:

- args

It is used in a function prototype to accept a varying number of arguments.

It's an iterable object.

Usage - def fun(*args)

- kwargs

It is used in a function prototype to accept the varying number of keyworded arguments.

·It's an iterable object

·Usage - def fun(**kwargs):

fun(colour="red".units=2)

27. "in Python, Functions Are First-class Objects." What Do You Infer from This?

Answer:

It means that a function can be treated just like an object. You can assign them to variables, or pass them as arguments to other functions. You can even return them from other functions.

28. What Is the Output Of: Print(__name__)? Justify Your Answer.

`__name__` is a special variable that holds the name of the current module. Program execution starts from main or code with 0 indentations. Thus, `__name__` has a value `__main__` in the above case. If the file is imported from another module, `__name__` holds the name of this module.

29. What Is a Numpy Array?

Answer:

A numpy array is a grid of values, all of the same type, and is indexed by a tuple of non-negative integers. The number of dimensions determines the rank of the array. The shape of an array is a tuple of integers giving the size of the array along each dimension.

30. What Is the Difference Between Matrices and Arrays?

Answer:

Matrices	Arrays
• A matrix comes from linear algebra and is a two-dimensional representation of data • It comes with a powerful set of mathematical operations that allow you to manipulate the data in interesting ways	• An array is a sequence of objects of similar data type • An array within another array forms a matrix

Advanced Level



31. How Do You Get Indices of N Maximum Values in a Numpy Array?

Answer:

```
>>import numpy as np
>>arr=np.array([1, 3, 2, 4, 5])
>>print(arr.argsort() [ -N: ][: : -1])
```

32. How Would You Obtain the Res_set from the Train_set and the Test_set from Below?

```
>>train_set=np.array([1, 2, 3])
>>test_set=np.array([[0, 1, 2], [1, 2, 3]])
Res_set  [[1, 2, 3], [0, 1, 2], [1, 2, 3]]
```

Choose the correct option:

- 1.res_set = train_set.append(test_set)
- 2.res_set = np.concatenate([train_set, test_set]))
- 3.resulting_set = np.vstack([train_set, test_set])
- 4.None of these



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Answer:

Here, options a and b would both do horizontal stacking, but we want vertical stacking. So, option c is the right statement.

```
resulting_set = np.vstack([train_set, test_set])
```

33. How Would You Import a Decision Tree Classifier in Sklearn? Choose the Correct Option.

1. `from sklearn.decision_tree import DecisionTreeClassifier`
2. `from sklearn.ensemble import DecisionTreeClassifier`
3. `from sklearn.tree import DecisionTreeClassifier`
4. None of these

Answer:

3. `from sklearn.tree import DecisionTreeClassifier`

34. You Have Uploaded the Dataset in Csv Format on Google Spreadsheet and Shared It Publicly. How Can You Access This in Python?

Answer:

We can use the following code:

```
>>link = https://docs.google.com/spreadsheets/d/...  
>>source = StringIO.StringIO(requests.get(link).content))  
>>data = pd.read_csv(source)
```

35. What Is the Difference Between the Two Data Series given Below?

`df['Name']` and `df.loc[:, 'Name']`, where:

```
df = pd.DataFrame(['aa', 'bb', 'xx', 'uu'], [21, 16, 50, 33], columns =  
['Name', 'Age'])
```

Choose the correct option:

1. 1 is the view of original dataframe and 2 is a copy of original dataframe
2. 2 is the view of original dataframe and 1 is a copy of original dataframe
3. Both are copies of original dataframe
4. Both are views of original dataframe

Answer:

3. Both are copies of the original dataframe.

36. You Get the Error “temp.Csv” While Trying to Read a File Using Pandas. Which of the Following Could Correct It?

Error:

Traceback (most recent call last): File "<input>", line 1, in<module> UnicodeEncodeError: 'ascii' codec can't encode character.

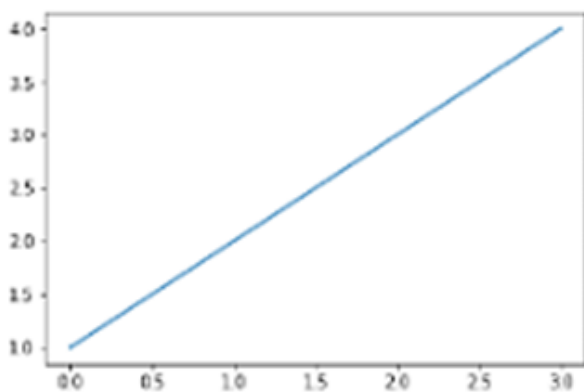
Choose the correct option:

- 1.pd.read_csv("temp.csv", compression='gzip')
- 2.pd.read_csv("temp.csv", dialect='str')
- 3.pd.read_csv("temp.csv", encoding='utf-8')
- 4.None of these

Answer:

The error relates to the difference between utf-8 coding and a Unicode. So option 3. pd.read_csv("temp.csv", encoding='utf-8') can correct it.

37. How Do You Set a Line Width in the Plot given Below?



```
>>import matplotlib.pyplot as plt
>>plt.plot([1,2,3,4])
>>plt.show()
```



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Choose the correct option:

- 1. `pd.read_csv("temp.csv", compression='gzip')`
- 2. `pd.read_csv("temp.csv", dialect='str')`
- 3. `pd.read_csv("temp.csv", encoding='utf-8')`
- 4. None of these

Answer:

- 3. In line two, write `plt.plot([1,2,3,4], lw=3)`

38. How Would You Reset the Index of a Dataframe to a given List?

Choose the Correct Option

- 1. `df.reset_index(new_index,)`
- 2. `df.reindex(new_index,)`
- 3. `df.reindex_like(new_index,)`
- 4. None of these

Answer:

- 3. `df.reindex_like(new_index,)`

39. How Can You Copy Objects in Python?

Answer:

The function used to copy objects in Python are:
`copy.copy` for shallow copy and
`copy.deepcopy()` for deep copy

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40. What Is the Difference Between range() and xrange() Functions in Python?

Answer:

range()	xrange()
• range returns a Python list object	• xrange returns an xrange object

41. How Can You Check Whether a Pandas Dataframe Is Empty or Not?

Answer:

The attribute df.empty is used to check whether a pandas data frame is empty or not.

```
>>import pandas as pd
>>df=pd.DataFrame({'A':[]})
>>df.empty
Output: True
```

42. If You Split Your Data into Train/Test Splits, Is It Possible to over Fit Your Model?

Answer:

Yes. One common beginner mistake is re-tuning a model or training new models with different parameters after seeing its performance on the test set.

43. Which Python Library Is Built on Top of Matplotlib and Pandas to Ease Data Plotting?

Answer:

Seaborn is a Python library built on top of matplotlib and pandas to ease data plotting. It is a data visualization library in Python that provides a high-level interface for drawing statistical informative graphs.

44. What are the important features of Python?

Answer:

- Python is a scripting language. Python, unlike other programming languages like C and its derivatives, does not require compilation prior to execution.
- Python is dynamically typed, which means you don't have to specify the kinds of variables when declaring them or anything.
- Python is well suited to object-oriented programming since it supports class definition, composition, and inheritance.

45. What type of language is Python?

Answer:

Although Python can be used to write scripts, it is primarily used as a general-purpose programming language.

46. Explain how Python is an interpreted language.

Answer:

Any programming language that is not in machine-level code before runtime is called an interpreted language. Python is thus an interpreted language.



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47. What is PEP 8?

Answer:

PEP denotes Python Enhancement Proposal. It's a collection of guidelines for formatting Python code for maximum readability.

48. Explain Python namespace.

Answer:

In Python, a namespace refers to the name that is assigned to each object.

49. What are decorators in Python?

Answer:

Decorators are used for changing the appearance of a function without changing its structure. Decorators are typically defined prior to the function they are enhancing.

50. How to use decorators in Python?

Answer:

Decorators are typically defined prior to the function they are enhancing. To use a decorator, we must first specify its function. Then we write the function to which it is applied, simply placing the decorator function above the function to which it must be applied.

51. Differentiate between .pyc and .py.

Answer:

The .py files are the source code files for Python. The bytecode of the python files are stored in .pyc files, which are created when code is imported from another source. The interpreter saves time by converting the source .py files to .pyc files.

52. What is slicing in Python?

Answer:

Slicing is a technique for gaining access to specific bits of sequences such as strings, tuples, and lists.



53. How to use the slicing operator in Python?

Answer:

Slicing is a technique for gaining access to specific bits of sequences such as lists, tuples, and strings. The slicing syntax is [start:end:step]. This step can also be skipped. [start:end] returns all sequence items from the start (inclusive) to the end-1 element. It means the ith element from the end of the start or end element is negative i. The step represents the jump or the number of components that must be skipped.

54. What are keywords in python?

Answer:

In Python, keywords are reserved words with a specific meaning. They are commonly used to specify the type of variables. Variable and function names cannot contain keywords. Following are the 33 keywords of Python:

- Yield
- For
- Else
- Elif
- If
- Not
- Or
- And
- Raise
- Nonlocal
- None
- Is
- In
- Import
- Global
- From
- Finally
- Except
- Del
- Continue
- Class
- Assert
- With
- Try
- False
- True
- Return
- Pass
- Lambda
- Def
- As
- Break
- While

55. How to combine dataframes in Pandas?

Answer:

The following are the ways through which the data frames in Pandas can be combined:

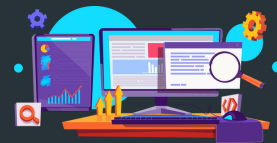
- Concatenating them by vertically stacking the two dataframes.
- Concatenating them by horizontally stacking the two dataframes.
- Putting them together in a single column.

56. What are the key features of the Python 3.9.0.0 version?

Answer:

- Zoneinfo and graphlib are two new modules.
- Improved modules such as asyncio and ast.
- Optimizations include improved idiom for assignment, signal handling, and Python built-ins.
- Removal of erroneous methods and functions.
- Instead of LLI, a new parser is based on PEG.
- Remove Prefixes and Suffixes with New String Methods.
- Generics with type hinting in standard collections.

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